Dynamic comparison of Monod and substrate inhibition models

Recombinant lipase production and oxygen transfer design

CL 420, Introduction to Biochemical Engineering

Final course project report



Group 21

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Abstract

A fed batch process for extracellular recombinant *Rhizopus oryzae* lipase production by *Pichia pastoris* is evaluated using Monod and substrate inhibition kinetics. Published rate functions and parameters are implemented in Python. Four material balances include nutrient feed volume and explicit methanol feeding policies. Eight reference simulations are followed by a constant substrate setpoint screen, oxygen transfer calculations, and a batch cost estimate. Under the inhibition model, a screened methanol setpoint of 3.4 g L^{-1} gives a predicted harvest activity of 321.35 U mL^{-1} after 28.13 h of induction. The corresponding productivity is 10,349 U L^{-1} h^{-1}. Oxygen enrichment and an independently qualified transfer capacity are necessary for this candidate operating point. The study connects kinetic model selection to feeding, oxygen supply, product recovery, and operating cost. All performance results are model predictions.

1. Product and process definition

1.1 Product, application and design basis

Recombinant *Rhizopus oryzae* lipase catalyses lipid hydrolysis and can support ester synthesis and transesterification. Its industrial applications include biodiesel and structured lipid production. The proposed product is a concentrated liquid enzyme preparation for industrial biocatalysis. The target customers are enzyme formulators and process developers [1, 2].

The design is based on a nominal 5 L stirred vessel and a 2 L initial induction volume. A 3.5 L working limit provides headspace for aeration and foam. Product output is expressed as lipase activity, with concentration in U mL^{-1} and total batch activity in U. Production and costs are evaluated per batch and per million recovered U.

1.2 Host and metabolic basis

The host is *Pichia pastoris* X-33 carrying pPICZαROL under the methanol induced P_{AOX1} promoter, with the Mut^+ phenotype. This strain secretes ROL into the broth, which allows cell removal without cell disruption. The host and cultivation conditions are adopted from the published ROL process [1].

Glycerol first supports biomass growth. After a transition phase, methanol acts as both carbon source and inducer. Alcohol oxidase converts methanol to formaldehyde and hydrogen peroxide. Catalase removes hydrogen peroxide. Formaldehyde is either assimilated through the xylulose monophosphate pathway or oxidized towards carbon dioxide for energy generation. Oxygen is required for methanol oxidation and respiration. Carbon dioxide and metabolic heat are important process outputs [1].

1.3 Critical quality attributes

Attribute	Proposed assessment
Enzyme activity	Activity per mL and total recovered U using a fixed assay.
Product integrity	Activity relative to protein content, with a proteolysis check.
Process impurities	Residual methanol, salts and viable host cells after recovery.
Storage stability	Retained activity during storage of the liquid preparation.

These are proposed release attributes. Numerical acceptance limits require application and stability data. Activity values retain the assay basis of the published kinetic parameters, a Roche colorimetric method on clarified culture liquid at 30 °C and pH 7.25, measured at 580 nm [1]. These conditions define the activity units used in the simulations.

2. Process configuration and operating basis

Fed batch operation controls residual methanol while avoiding continuous product withdrawal. It also separates biomass development from induction. Excess methanol can inhibit growth and production, while insufficient methanol limits the induced process. This supports controlled feeding rather than a large initial methanol charge [3].

Table 1: Operating basis. L denotes literature input and D denotes a design assumption.

Quantity	Value or operating rule	Basis
Vessel	Stirred tank, nominal 5 L	L [1]
Induction start	$V_0 = 2$ L, $X_0 = 27$ g L^{-1}, $P_0 = 50$ U mL^{-1}	L [1]
Residual methanol	2 g L^{-1} for all reference starts	D
Temperature and pH	30 °C and 5.5	L [1]
Dissolved oxygen	25% of air saturation	L [1]
Agitation and gas	800 to 1000 rpm, 1 to 2 vvm with oxygen enrichment	L [1]
Working limits	3.5 L, 55 g L^{-1} biomass, 10 g L^{-1} methanol	D, L [1]

The proposed 0.13 m tank diameter gives a 151 mm liquid depth at 2 L. Impeller centres at 40 and 105 mm, with 13 mm blades, keep both turbines submerged initially.

2.1 Inoculum, medium and process sequence

The proposed inoculum stage adopts YPD medium in baffled flasks for 24 h at 30 °C and 150 rpm. Cells are collected and resuspended in basal medium. The seed stage can run in parallel with reactor preparation [1].

The basal medium contains, per litre, 40 g glycerol, 26.7 mL of 85% phosphoric acid, 0.93 g $CaSO_4$, 18.2 g K_2SO_4, 14.9 g $MgSO_4 7H_2O$, 4.13 g KOH, 5 mL trace stock, 2 mL biotin stock and 0.5 mL antifoam. Trace stock supplies Fe, Zn, Cu, Mn, Co and other micronutrients [1].

After glycerol depletion, the design adopts a 5 h transition with glycerol feeding and stepwise methanol addition. Induction then uses methanol with trace nutrients and a linked 200 g L^{-1} NH_4Cl feed. The ratio is 0.12 g NH_4Cl per g methanol [1]. The simulation begins at induction, not at inoculation. The scheduling allowance is 18 h for the glycerol batch, 5 h for transition, 28.13 h for induction and 6 h for turnaround. The transition duration is literature based. The resulting reactor occupancy is 57.13 h.

3. Growth and product formation kinetics

The inhibition model uses substrate dependent growth and production rates. The Monod comparator removes the inhibition term while retaining the same nominal coefficients [1].

$$q_i^{\mathrm{H}}(S) = q_{\max,i}\frac{S}{K_{S,i} + S + S^2/K_{I,i}}, \tag{1}$$

$$q_i^{\mathrm{M}}(S) = q_{\max,i}\frac{S}{K_{S,i} + S}, \qquad i = X, P. \tag{2}$$

Here $q_X = \mu$. This comparison tests the effect of a modelling assumption. It is not a separate fit of Monod parameters to experimental data.

Table 2: Kinetic inputs from the thesis, Table 5.1 [1].

Parameter	Value	Unit
$\mu_{\max}$, $K_{S,X}$, $K_{I,X}$	0.069, 0.50, 7.5	$\mathrm{h^{-1}}$, $\mathrm{g\ L^{-1}}$, $\mathrm{g\ L^{-1}}$
$q_{\max,P}$, $K_{S,P}$, $K_{I,P}$	2200, 10.0, 1.2	$\mathrm{U\ g^{-1}\ h^{-1}}$, $\mathrm{g\ L^{-1}}$, $\mathrm{g\ L^{-1}}$
$Y_{S/X}$, m_S	5.14, 0.013	$\mathrm{g\ g^{-1}}$, $\mathrm{g\ g^{-1}\ h^{-1}}$
$Y_{O_2/X}$, m_{O_2}	0.206, 6.1×10^{-4}	$\mathrm{mol\ g^{-1}}$, $\mathrm{mol\ g^{-1}\ h^{-1}}$

$$q_S = 5.14\mu + 0.013, \qquad q_{O_2} = 0.206\mu + 0.00061. \tag{3}$$

The units are g methanol $\mathrm{g^{-1}\ h^{-1}}$ and mol $\mathrm{O_2\ g^{-1}\ h^{-1}}$, respectively. The coefficient 5.14 is substrate per biomass. Its reciprocal, 0.1946 g $\mathrm{g^{-1}}$, is the intrinsic biomass yield before maintenance.

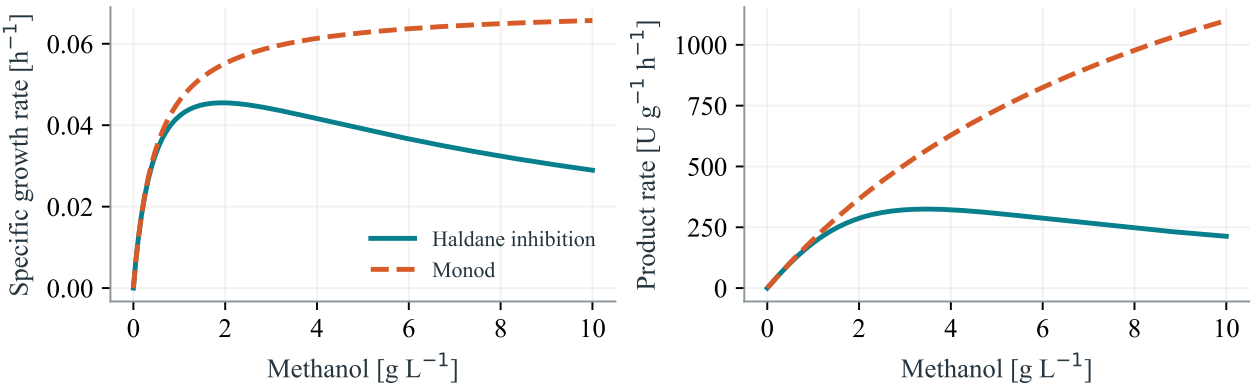


Figure 1: Rate laws used in the study. Inhibition produces finite substrate optima.

The inhibition peaks occur at $S = \sqrt{K_S K_I}$, about 1.94 g $\mathrm{L^{-1}}$ for growth and 3.46 g $\mathrm{L^{-1}}$ for production. The achieved peak rates are 0.0455 $\mathrm{h^{-1}}$ and 324.8 U $\mathrm{g^{-1}\ h^{-1}}$, below the nominal coefficients. Production is modelled as growth linked during methanol induction, with a separate $q_P(S)$ law and a variable activity to biomass yield. This is not a constant q_P/μ relationship. Production in nongrowing cells is not established by this formulation.

4. Material balances and numerical method

Let $B = XV$, $M = SV$ and $A = 1000PV$ denote total biomass in g, methanol in g and product activity in U. The factor 1000 converts P from U mL^{-1} to U L^{-1}. The four states are V, B, M, A. Methanol feed F is in g h^{-1}.

$$\frac{dV}{dt} = aF, \qquad a = \frac{1}{792} + \frac{0.12}{200} = 0.0018626\,\mathrm{L\,g^{-1}}, \tag{4}$$

$$\frac{dB}{dt} = \mu B, \qquad \frac{dM}{dt} = F - q_S B, \tag{5}$$

$$\frac{dA}{dt} = q_P B, \qquad X = B/V, \quad S = M/V. \tag{6}$$

The additive liquid volume approximation uses a 20 °C methanol feed density of 792 g L^{-1} [10] and the linked nitrogen stock [1]. Sampling, evaporation, base and antifoam additions are omitted. A 5% working volume allowance is checked separately. Amount balances avoid ambiguity from changing volume, consistent with the course fed batch treatment [8].

Table 3: Feeding policies implemented in this study.

Case	Definition
CF	$F = 14$ g h^{-1}.
CM	$F = q_S(0.03)B_0 \exp(0.03t)$. A prescribed feed, not biomass feedback.
LM	$F = q_S(\mu_T)B_0 \exp(\int_0^t \mu_T\, dt)$, with $\mu_T = 0.01 + 0.0005t$ up to 60 h, then 0.04 h^{-1}.
CS	$F = q_S B/(1 - aS)$, the ideal balance required for constant S.

CF, CM and CS target values follow the published strategy definitions. The 60 h LM ramp is an explicit study assumption. CM and LM targets need not equal the actual transient growth rate. Their feed histories are identical across the two kinetic models. CS uses each model's uptake rate, so its feed history differs between models. It represents an ideal setpoint policy, rather than a simulated physical controller.

Classical fourth order Runge Kutta uses a 0.01 h step. Integration stops at the first biomass, volume, methanol or 120 h limit. Event times are refined within the final step. All eight cases pass step halving, with maximum relative harvest activity change below 3×10^{-10}. Integrated substrate and activity balances are also checked. An independent closed form CS solution verifies the selected case to below 10^{-12} relative error. These are numerical checks, not experimental validation.

5. Dynamic comparison and reference results

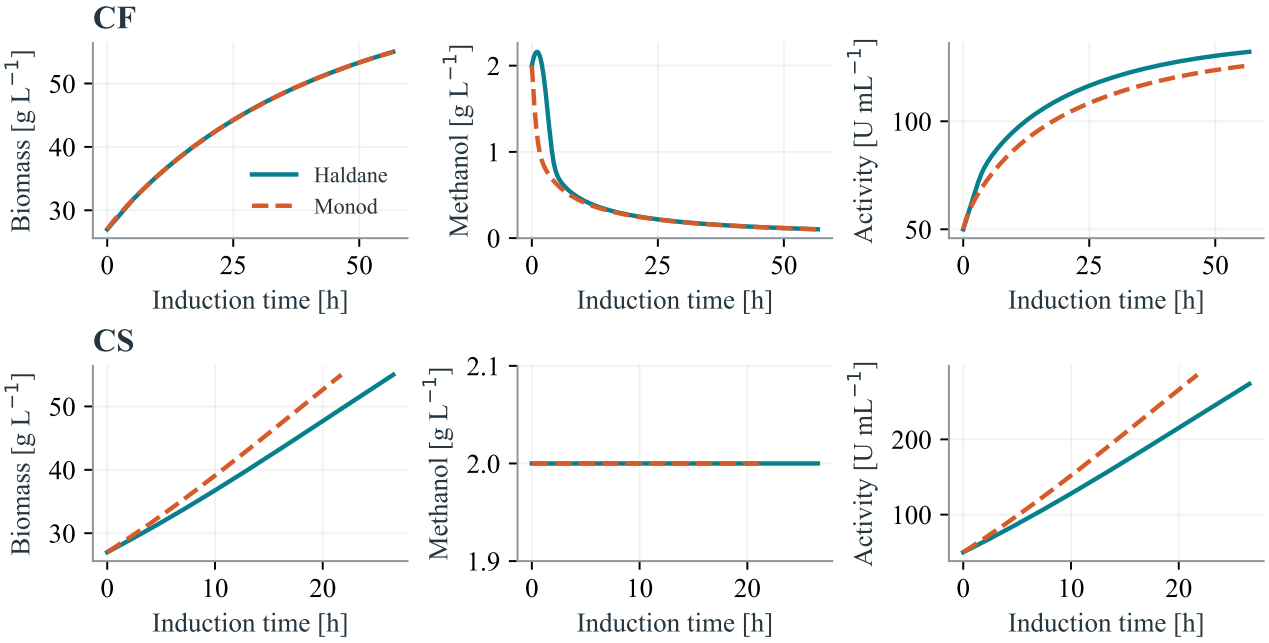


Figure 2: Reference CF and CS trajectories. Each model pair has the same initial state and operating limits. CS holds 2 g L^{-1}.

Table 4: Kinetic reference endpoints before oxygen screening. Each run reaches 55 g L^{-1} biomass.

Model	Feed	t_f h	V_f L	P_f U mL^{-1}	Q_P U L^{-1} h^{-1}	Peak OUR mol h^{-1}
Haldane	CF	56.74	3.480	132.15	1,823	0.614
Haldane	CM	40.76	3.344	139.03	2,677	1.245
Haldane	LM	53.11	3.385	136.79	2,020	1.509
Haldane	CS	26.55	3.285	273.67	9,162	1.803
Monod	CF	56.75	3.480	125.85	1,711	0.647
Monod	CM	40.76	3.344	135.96	2,602	1.245
Monod	LM	53.10	3.384	133.57	1,959	1.509
Monod	CS	21.73	3.258	285.94	11,746	2.147

Net volumetric productivity is defined as $Q_P = (A_f - A_0)/(V_f t_f)$. The inhibition CS reference gives 273.67 U mL^{-1} and 9,162 U L^{-1} h^{-1}. Monod CS gives 285.94 U mL^{-1} and a shorter induction time. Under CF, Monod instead predicts a lower harvest activity. The effect of removing inhibition depends on the resulting substrate and biomass trajectory, not only the instantaneous rate curve.

These references assume sufficient oxygen and exclude minor feed volume allowances. All starts use 2 g L^{-1} methanol. The selected design is checked separately for the 5% volume reserve and oxygen constraint.

6. Oxygen transfer and reactor design

Derived demand is compared with gas transfer capacity at a fixed dissolved oxygen target [1, 8].

$$\mathrm{OUR}_V = q_{O_2} X, \qquad \mathrm{OTR}_V = k_L a \, \Delta C_{\mathrm{lm}}. \tag{7}$$

The adopted air saturation value at 30 °C is $C^*_{\mathrm{air}} = 2.3 \times 10^{-4}$ mol L^{-1} [1]. The 25% air calibrated DO target gives $C_L = 5.75 \times 10^{-5}$ mol L^{-1}. Oxygen enrichment changes C^*, while this absolute DO target is retained.

Gas flow is specified at 30 °C and 1 bar absolute. One mol CO_2 is assumed to replace each mol O_2 consumed, keeping dry gas flow constant. Let $g = G/V$, $c = C^*_{\mathrm{air}}/0.21$ and y denote oxygen mole fraction. For plug flow gas and well mixed liquid,

$$G = \frac{60\, q_g V\, p}{RT}, \qquad y_{\mathrm{out}} = y_{\mathrm{in}} - \frac{\mathrm{OUR}}{G}, \tag{8}$$

$$\mathrm{OTR}_{V,\mathrm{max}} = \frac{g}{c}(c y_{\mathrm{in}} - C_L)\left[1 - \exp\left(-\frac{k_L a c}{g}\right)\right]. \tag{9}$$

Equation 9 integrates the gas balance and includes oxygen depletion at the capacity itself. At fixed vvm, gas blend, DO and $k_L a$, volumetric capacity is constant. Pressure gradients and broth solubility corrections are omitted.

Table 5: Proposed geometry and transfer qualification basis. Geometry and power factors are design assumptions.

Quantity	Selected value or estimate
Tank and impeller diameters	0.130 m and 0.065 m, two Rushton turbines
Agitation and total gas	1000 rpm and 2 vvm, within literature ranges
Power number and gassing factor	$N_P = 5.0$, $P_g/P_0 = 0.70$
Calculated power	$P_0 = 53.7$ W, $P_g = 37.6$ W
Harvest power and gas velocity	11.39 kW m^{-3} and 0.00829 m s^{-1}
Qualification requirement	$k_L a \geq 1000$ h^{-1} in the actual broth

The transfer correlations use SI units, with P_g/V in W m^{-3}, u_s in m s^{-1} and $k_L a$ in s^{-1} [4, 6, 7].

$$\begin{aligned} k_L a_{\mathrm{water}} &= 0.026 (P_g/V)^{0.4} u_s^{0.5}, \\ k_L a_{\mathrm{electrolyte}} &= 0.002 (P_g/V)^{0.7} u_s^{0.2}. \end{aligned} \tag{10}$$

Power follows $P_0 = n_i N_P \rho N^3 D_i^5$, with $N = 1000/60$ s^{-1} and $\rho = 1000$ kg m^{-3}. Correlations give 357 and 1908 h^{-1} for water and electrolyte [4, 6, 7]. These are screening extrapolations. The 1000 h^{-1} target requires broth measurement.

7. Oxygen feasibility and feeding selection

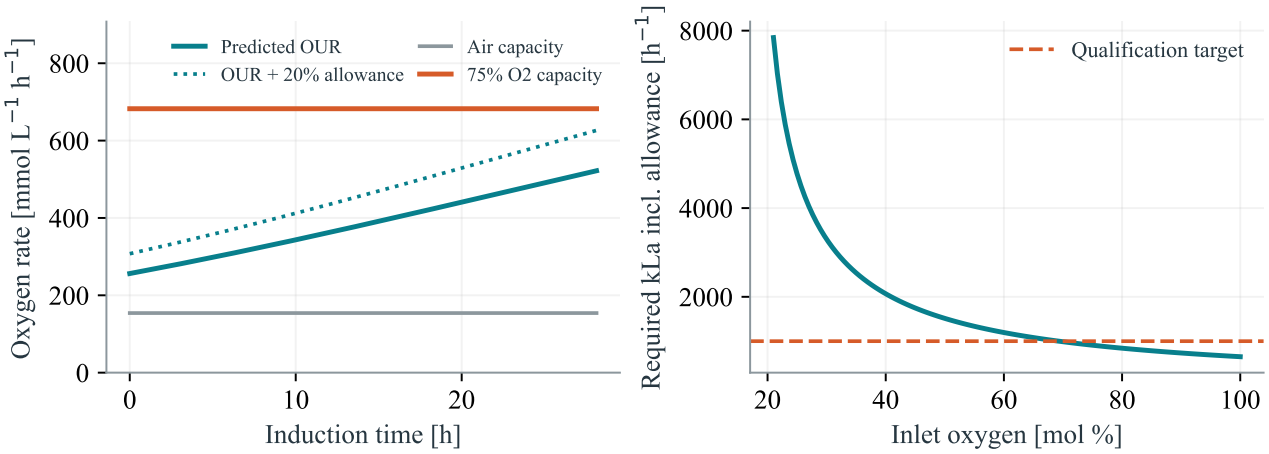


Figure 3: Transfer screen for the selected inhibition case at 3.4 g L^{-1}. Capacity uses the 1000 h^{-1} qualification target and the stated gas balance.

At harvest, $OUR_V = 522.0$ mmol L^{-1} h^{-1} and total OUR is 1.723 mol h^{-1}. At 75% inlet O_2 and 2 vvm, $H = (g/c)(cy_{in} - C_L) = 3.3207$ mol L^{-1} h^{-1}. Inverting Equation 9 gives

$$k_L a_{req} = -\frac{4.7608}{0.00109524} \ln\left[1 - \frac{1.20(0.52204)}{3.3207}\right] \approx 909\,h^{-1}.$$

At the 1000 h^{-1} qualification target, capacity is 682.4 mmol L^{-1} h^{-1} and exceeds demand by at least 20%. Air alone fails this screen. The DO loop adjusts agitation and enrichment.

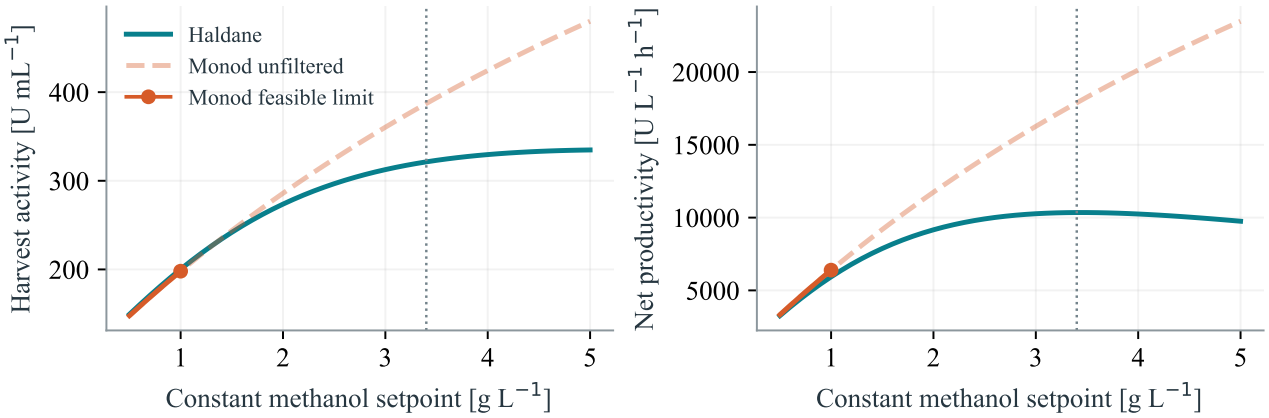


Figure 4: Constant setpoint screen in 0.1 g L^{-1} increments. Each case starts at its own setpoint. Faded Monod portions fail the oxygen screen. The vertical line marks the selected inhibition case.

Under inhibition, the best feasible grid point is 3.4 g L^{-1}, giving 321.35 U mL^{-1} and 10,349 U L^{-1} h^{-1}. Productivity is 12.9% above the 2 g L^{-1} reference. Setpoints from 3.2 to 3.6 g L^{-1} differ by less than 0.3%. Under the same oxygen screen, Monod's best feasible point is 1.0 g L^{-1} and 6,391 U L^{-1} h^{-1}, despite its rising unfiltered curve. These are candidate points, not proven biological optima, and exclude controller transients.

8. Sensitivity, monitoring and control

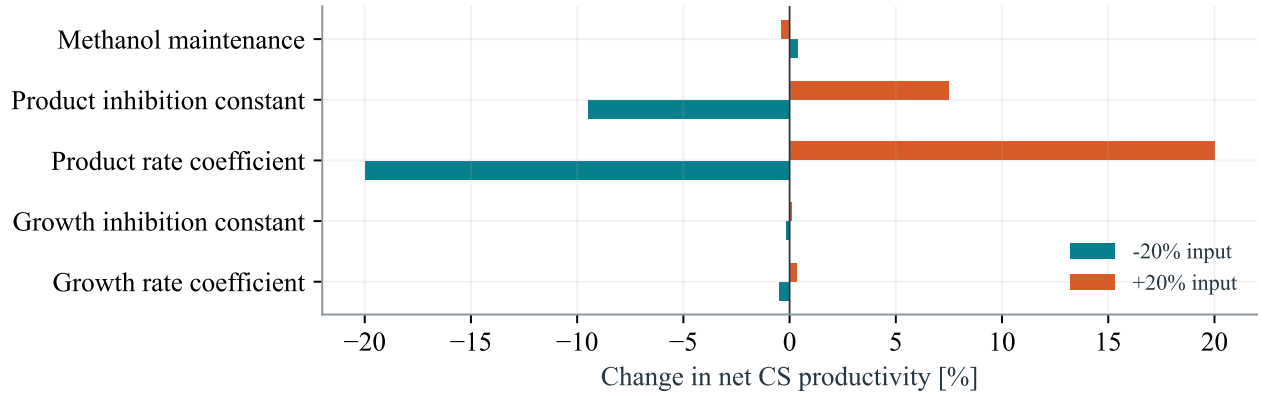


Figure 5: One input at a time sensitivity around the selected inhibition case. Each listed coefficient is changed by 20%, while the setpoint and remaining inputs are held fixed.

The scenarios are not confidence intervals. Large product curve coefficient uncertainty limits setpoint precision [1]. At constant substrate, faster growth shortens induction but also reduces accumulated activity. These effects largely cancel in Q_P, explaining its weak growth sensitivity. Product rate uncertainty directly changes activity and cost. Increasing the growth coefficient by 20% violates the oxygen allowance despite little change in Q_P.

Table 6: Proposed process analytical technology and control actions. Published monitoring methods are extended into a design control plan [1].

Variable	Measurement	Action or interpretation
Methanol	Online sensor, HPLC check	Adjust feed to the chosen setpoint.
DO and gas	DO probe, inlet flow, off gas O_2/CO_2	Cascade agitation and oxygen blend. Reduce feed if capacity is exhausted.
pH and temperature	Online probes	KOH dosing and jacket cooling.
Volume and foam	Feed scales, level and foam probes	Enforce working limit and antifoam additions.
Biomass and nutrients	Offline DCW and ammonium	Check growth and linked nutrient supply.
Product and viability	Offline activity, protein and viable cells	Determine harvest readiness and detect degradation.

The methanol controller implementation is not modelled. An operating system would require sensor calibration, feed pump limits and tested tuning. High agitation is supported by the published stirred process, but strain specific shear limits and broth rheology remain unmeasured. Cell viability, foam and protease release should be checked when increasing agitation. The transfer correlations assume a low viscosity liquid and do not resolve these effects.

9. Recovery and rough batch economics

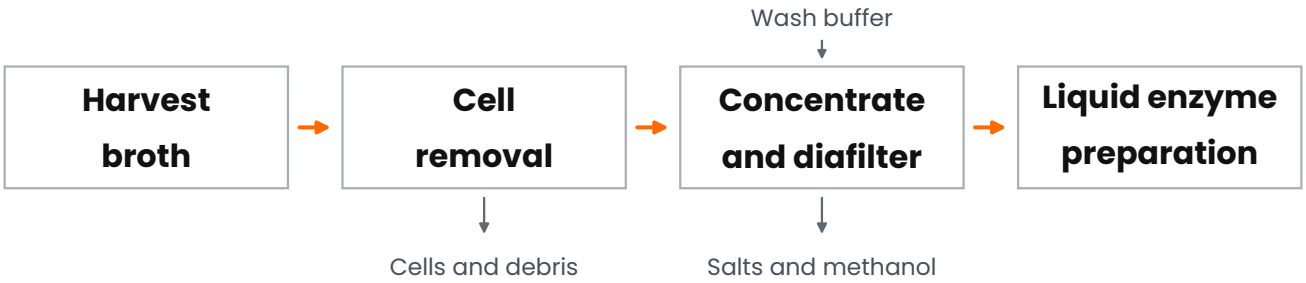


Figure 6: Proposed recovery sequence. Lipase remains in the liquid product stream.

Published ROL processing uses centrifugation, microfiltration and ultrafiltration [2]. Diafiltration is proposed for methanol and salt removal, subject to retention and activity checks. Chromatography and drying are omitted because the selected product is an industrial liquid preparation. A combined 70% activity recovery is assumed, with 50% and 90% scenarios. Membrane retention, product stability and actual purity require confirmation.

The budget below uses explicit planning rates in INR, not supplier quotations. Reactor occupancy includes the assumed preinduction and turnaround periods. Equipment use is charged through an occupancy rate. Capital purchase, taxes, sales margins and a selling price are outside this estimate. All gas volumes use 30 °C and 1 bar.

Table 7: Illustrative cost for the selected batch. Detailed quantities are calculated by the accompanying Python model.

Cost group	Assumed rate or allowance	INR
Medium and feeds	Seed/salts 450, transition/minor stocks 250, glycerol 200/kg, methanol 80/kg, NH_4Cl 200/kg	789
Gas and utilities	O_2 100/m^3, air 10/m^3, electricity 10/kWh, thermal allowance 300	964
Reactor occupancy	57.13 h at INR 100/h	5,713
Recovery	Membrane use, clarification and buffers	1,500
Labour	6 h at INR 150/h	900
Waste treatment	Batch allowance	150
Total	Rounded sum from unrounded calculations	10,016

Recovered activity is $0.70 \times 1.061 \times 10^6 = 0.743 \times 10^6$ U. The corresponding cost is INR 13,487 per million recovered U. At 50% and 90% recovery, it becomes INR 18,882 and INR 10,490, respectively. A separate 25% change in all budget rates changes total cost by 25%. This is a preliminary batch budget, not an industrial profitability claim.

10. Design summary and scale up considerations

Table 8: Selected inhibition model operating point and engineering outputs.

Quantity	Predicted value or requirement
Methanol setpoint and induction time	3.4 g L^{-1} and 28.13 h
Harvest volume and biomass	3.301 L and 181.56 g DCW
Methanol and linked NH_4Cl feed	698.57 g and 83.83 g
Pure methanol and nitrogen stock volumes	0.882 L and 0.419 L
Peak methanol feed	42.87 g h^{-1}
Harvest activity and net productivity	321.35 U mL^{-1} and 10,349 U L^{-1} h^{-1}
Predicted $Y_{X/S}$ and activity $Y_{P/S}$	0.1838 g g^{-1} and 1,384 U g^{-1}
Peak OUR and transfer target	1.723 mol h^{-1} and $k_La \geq 1000$ h^{-1}
Total inlet gas at harvest	6.60 L min^{-1}, 75% O_2 upper capacity case

For example, $\Delta B = 55(3.301) - 27(2) \approx 127.56$ g and methanol consumption is 694.15 g. Thus $Y_{X/S} = 127.56/694.15 \approx 0.1838$ g g^{-1}. The activity yield uses net induction activity, $Y_{P/S} = 0.961 \times 10^6/694.15 \approx 1,384$ U g^{-1}. These yields include maintenance. Adding 5% to the selected working volume gives 3.466 L, below the 3.5 L limit. Net productivity over the assumed 57.13 h reactor cycle is 5,095 U L^{-1} h^{-1}, compared with 10,349 during induction alone.

10.1 Industrial scale up

Industrial transfer requires matching oxygen supply, mixing and heat removal, rather than holding rpm or vvm unchanged. Ponte reports poorer ROL performance after transfer to 50 L despite nominal DO control, showing that a probe reading alone does not establish equivalent cell conditions. A separate *Pichia pastoris* study demonstrated 1000 L operation using pressure and a revised methanol feed, but produced a different intracellular enzyme. It supports pressure and feeding as scale up options, not transfer of its titer to ROL [1, 5]. A pilot study should first establish broth k_La, mixing time, off gas depletion, cooling duty and activity recovery. No industrial production rate is extrapolated from the present bench model.

10.2 Conclusion and limits

The inhibition model supports a candidate constant substrate operating region near the product rate peak. Its best screened point improves nominal productivity while requiring enriched oxygen and verified transfer capacity. The Monod comparator favours a different substrate policy. The decision is conditional on the kinetic inputs and the stated equipment qualification. The model excludes adaptation lag, oxygen limited physiology, degradation and spatial gradients. Its four reactor states do not make it an intracellular structured model [9]. Experimental kinetic checks and broth transfer measurements are required before operation.

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